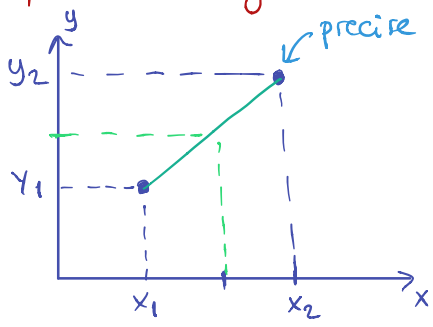
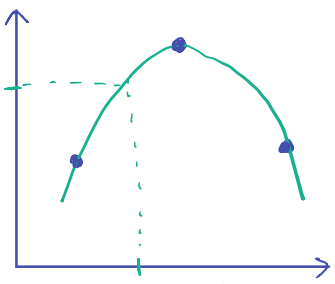
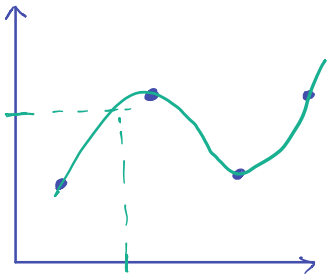
Chapter 17 – Polynomial Interpolation



line : 1st order polynomial



A parabola



Goal : to estimate intermediate data pts between precise data points.

$(n-1)^{th}$ order polynomial

$$f(x) = a_1 + a_2 x + a_3 x^2 + \cdots + a_n x^{n-1}$$

Polynomial interpolation

n data pts are needed to determine n coefficients.

Ex Data: $x_1 = 300$, $f(x_1) = 0.616$

$x_2 = 400$, $f(x_2) = 0.525$

$x_3 = 500$, $f(x_3) = 0.457$

want to determine the coefficients of the parabola $f(x) = p_1 x^2 + p_2 x + p_3$ that passes through the points above

Soln : Put each pt. into the equation:

$$\left.\begin{aligned} 0.616 &= p_1(300)^2 + p_2(300) + p_3 \\ 0.525 &= p_1(400)^2 + p_2(400) + p_3 \\ 0.457 &= p_1(500)^2 + p_2(500) + p_3 \end{aligned}\right\} \text{3 eqns, 3 unknowns}$$

in matrix form

$$\begin{bmatrix} 90000 & 300 & 1 \\ 160000 & 400 & 1 \\ 250000 & 500 & 1 \end{bmatrix} \begin{bmatrix} p_1 \\ p_2 \\ p_3 \end{bmatrix} = \begin{bmatrix} 0.616 \\ 0.525 \\ 0.457 \end{bmatrix} \Rightarrow$$

Solve:

$p_1 = 0.00\,000115$

$p_2 = -0.001715$

$p_3 = 1.027$

$$f(x) = 0.0000115x^2 - 0.001715x + 1.027$$

Evaluate $f(x)$ at an intermediate value:

$$f(350) = 0.567625$$

Drawbacks of above approach:

Coefficient matrix:

$$\underbrace{\begin{bmatrix} x_1^2 & x_1 & 1 \\ x_2^2 & x_2 & 1 \\ x_3^2 & x_3 & 1 \end{bmatrix}}_{\text{very ill-conditioned}} \begin{bmatrix} p_1 \\ p_2 \\ p_3 \end{bmatrix} = \begin{bmatrix} f(x_1) \\ f(x_2) \\ f(x_3) \end{bmatrix}$$

$\text{cond}(A) = 5.89 \times 10^6$ (for prev. exercise)

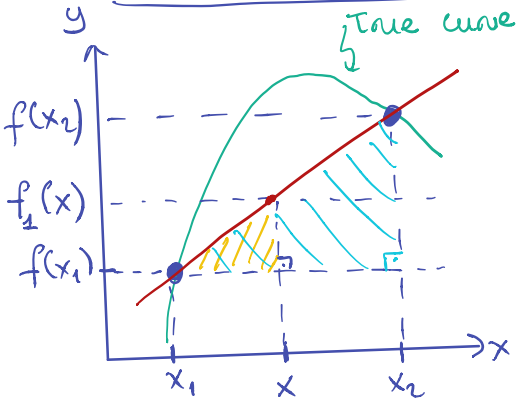
Means: 6 digits of the solution is questionable

* Ill-conditioning becomes worse if # eq^{n}s are bigger
* Approaches exist, which do not have this drawback

Newton Interpolating Polynomial

An alternative form of expressing an interpolating polynomial

Linear Interpolation: connect 2 pts with a straight line



using similar triangles (yellow & blue)

1st order

$$\frac{f_1(x) - f(x_1)}{x - x_1} = \frac{f(x_2) - f(x_1)}{x_2 - x_1}$$

Arrange:

slope of line

$$f_1(x) = f(x_1) + \frac{f(x_2) - f(x_1)}{x_2 - x_1}(x - x_1)$$

Newton interpolation formula

* Given 2 pts: $(x_1, f(x_1))$, $(x_2, f(x_2))$

* The smaller the interval btwn data pts, the better the approximation.

EX Estimate $\ln 2$ using linear interpolation

(a) Interpolate btwn $\ln 1$ and $\ln 6 = 1.791759$

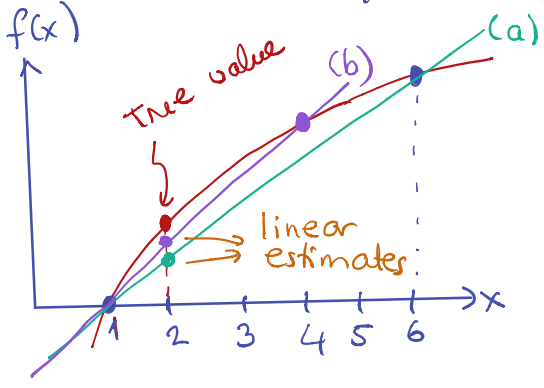
(b) " " $\ln 1$ and $\ln 4 = 1.386294$

(Note true value is $\ln 2 = 0.6931472$)

$$f_1(x) = f(x_1) + \frac{f(x_2) - f(x_1)}{x_2 - x_1}(x - x_1)$$

(a) $\ln 2 = \ln 1 + \frac{\ln 6 - \ln 1}{6 - 1}(2 - 1) = 0.358 \Rightarrow \varepsilon_t = 48.3\%$

(b) $\ln 2 = \ln 1 + \frac{\ln 4 - \ln 1}{4 - 1}(x - 1) = 0.4620 \Rightarrow \varepsilon_t = 33\%$



Quadratic Interpolation

If 3 data pts are available, we can fit a 2nd order polynomial

available pts:
$(x_1, f(x_1))$
$(x_2, f(x_2))$
$(x_3, f(x_3))$

(1) $$\boxed{f_2(x) = b_1 + b_2(x-x_1) + b_3(x-x_1)(x-x_2)}$$

when $x = x_1 \Rightarrow$ $\boxed{f(x_1) = b_1}$ (2)

when $x = x_2 \Rightarrow$ $f(x_2) = f(x_1) + b_2(x_2 - x_1)$

$$\boxed{b_2 = \frac{f(x_2) - f(x_1)}{(x_2 - x_1)}} \quad (3)$$

(slope of line connecting pts x_1 and x_2)

when $x = x_3 \Rightarrow$ $f(x_3) = \underbrace{f(x_1)}_{b_1} + \underbrace{\frac{f(x_2) - f(x_1)}{x_2 - x_1}}_{b_2}(x_3 - x_1) + b_3(x_3 - x_1)(x_3 - x_2)$

↓ solve for b_3

$$\Rightarrow \quad b_3 = \frac{\dfrac{f(x_3) - f(x_2)}{x_3 - x_2} - \dfrac{f(x_2) - f(x_1)}{x_2 - x_1}}{x_3 - x_1} \quad (4)$$

↓ similar to the 2nd order derivative formula

* Eq (1) has a structure similar to Taylor expansion.

Ex Approximate $\ln 2$ using a 2nd order polynomial, 3 pts:

$x_1 = 1$, $f(x_1) = 0$

$x_2 = 4$, $f(x_2) = 1.386294$

$x_3 = 6$, $f(x_3) = 1.791759$

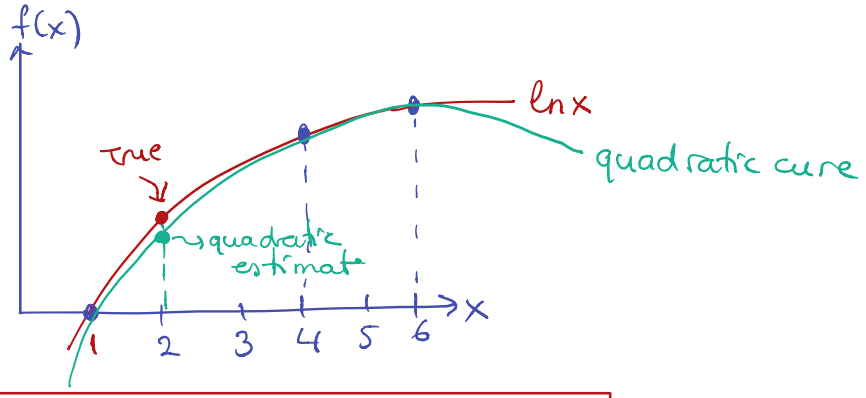
From (2) $b_1 = 0$

From (3) $b_2 = 0.462$

From (4) $b_3 = -0.0518$

$$f_2(x) = \underbrace{0.462098}_{b_2}(x-1) - \underbrace{0.0518}_{b_3}(x-1)(x-4)$$

$$f_2(2) = 0.5658 \quad \Rightarrow \varepsilon_t = 18\%$$



Lagrange Interpolating Polynomial

Formulate the linear interpolating polynomial as the weighted average of two values:

Given: $(x_1, f(x_1))$, $(x_2, f(x_2))$

$$f(x) = L_1 f(x_1) + L_2 f(x_2) \qquad (1)$$

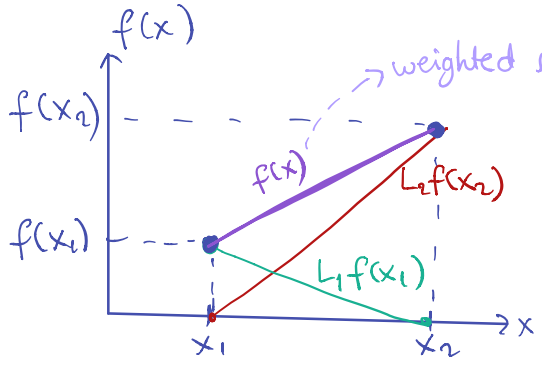
$$L_1 = \frac{x-x_2}{x_1-x_2} : \begin{cases} 1, \text{ at } x = x_1 \\ 0, \text{ at } x = x_2 \end{cases} \qquad (2)$$

$$L_2 = \frac{x-x_1}{x_2-x_1} : \begin{cases} 0, \text{ at } x_1 \\ 1, \text{ at } x_2 \end{cases} \qquad (3)$$

Put (2) and (3) into (1):

$$f(x) = \frac{x-x_2}{x_1-x_2} f(x_1) + \frac{x-x_1}{x_2-x_1} f(x_2)$$

Linear Lagrange interpolating polynomial



A parabola through three points:

Given: 3 pts: $(x_1, f(x_1))$, $(x_2, f(x_2))$, $(x_3, f(x_3))$
(pt 1, pt 2, pt 3)

Use 3 parabolas with each one passing thru one of the pts and is equal to zero at the other two.

2nd order Lagrange interpolating polynomial is:

$$f_2(x) = \underbrace{\frac{(x-x_2)(x-x_3)}{(x_1-x_2)(x_1-x_3)}}_{L_1(x)} f(x_1) + \underbrace{\frac{(x-x_1)(x-x_3)}{(x_2-x_1)(x_2-x_3)}}_{L_2(x)} f(x_2) + \underbrace{\frac{(x-x_1)(x-x_2)}{(x_3-x_1)(x_3-x_2)}}_{L_3(x)} f(x_3)$$

$L_1(x)$: zero at x_2 and x_3, one at x_1

$L_2(x)$: zero at x_1, x_3, one at x_2

$L_3(x)$: zero at x_1, x_2, one at x_3

General formula:

$$f_{n-1}(x) = \sum_{i=1}^{n} L_i(x) f(x_i) \quad \text{where} \quad L_i(x) = \prod_{\substack{j=1 \\ j \neq i}}^{n} \frac{x - x_j}{x_i - x_j}$$

Ex Given the measurements

$x_1 = 0$, $f(x_1) = 3.85$

$x_2 = 20$, $f(x_2) = 0.8$

$x_3 = 40$, $f(x_3) = 0.212$

Evaluate the function at $T = 15^\circ C$.

Use Lagrangian interpolation: a) First order polynomial
b) 2nd " "

} Self-study exercise

Soln First order polynomial

$$f_1(x) = \frac{(x-x_2)}{(x_1-x_2)} f(x_1) + \frac{(x-x_1)}{(x_2-x_1)} f(x_2)$$

$$f_1(x) = \frac{(x-20)}{(0-20)}(3.85) + \frac{(x-0)}{(20-0)}(0.8)$$

$$f_1(15) = 1.5625$$

Second order polynomial:

$$f_2(x) = \frac{(x-x_2)(x-x_3)}{(x_1-x_2)(x_1-x_3)} f(x_1) + \frac{(x-x_1)(x-x_3)}{(x_2-x_1)(x_2-x_3)} f(x_2) + \frac{(x-x_1)(x-x_2)}{(x_3-x_1)(x_3-x_2)} f(x_3)$$

$$f_2(x) = \frac{(x-20)(x-40)}{(0-20)(0-40)}(3.85) + \frac{(x-0)(x-40)}{(20-0)(20-40)}(0.8) + \frac{(x-0)(x-20)}{(40-0)(40-20)} 0.212$$

$$f_2(15) = 1.3316$$